# Lecture 14: Synthesis
# Solar System in Context, Astrobiology

## Planetary Systems


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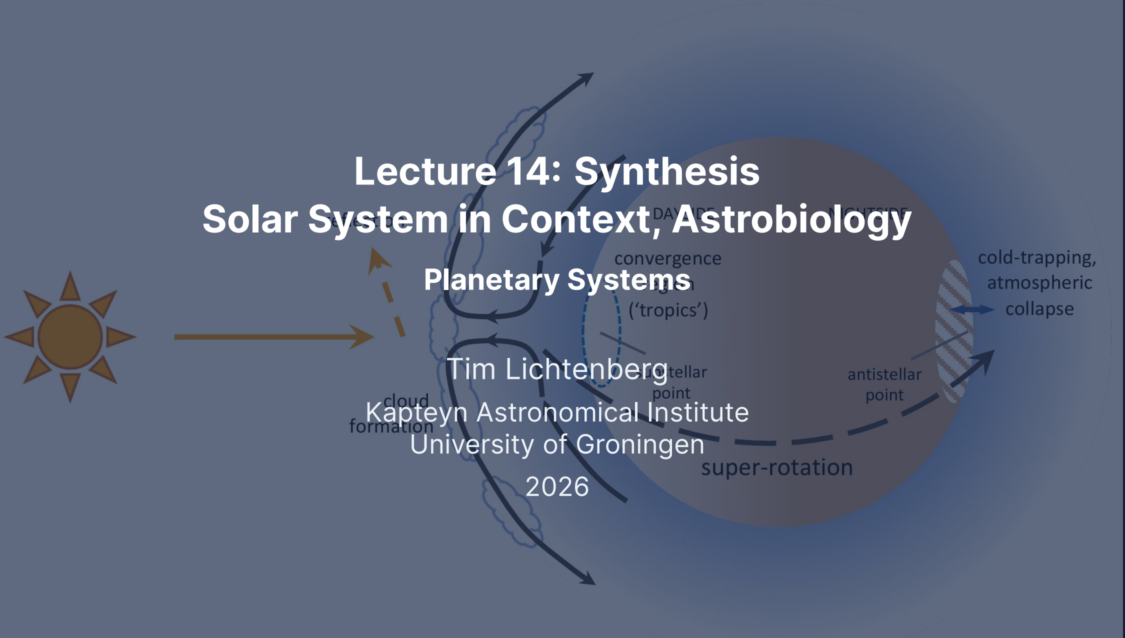


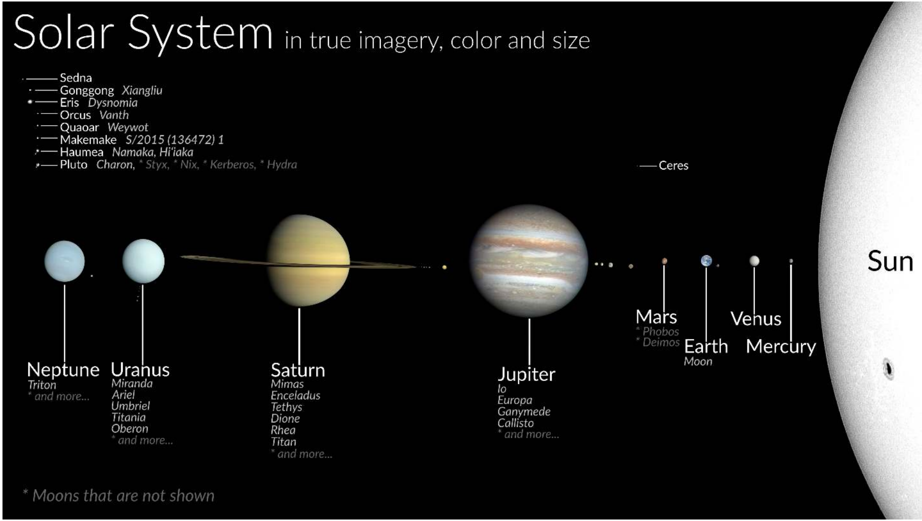

Tidally locked rocky planet atmospheric circulation. Wordsworth & Kreidberg (2022)

# Learning Objectives

By the end of this lecture, you will be able to:

- Place our solar system within the observed exoplanet population
- Evaluate habitability as a coupled *systems* property and derive the classical habitable-zone boundaries
- Identify the most promising life-detection targets for the next decade

Habitability is a trajectory, not a snapshot.

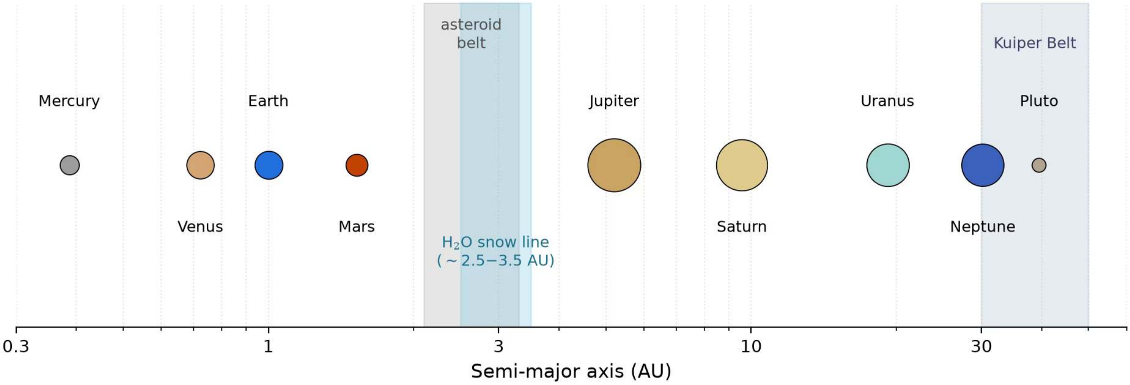



The principal bodies of the solar system grouped by region at true relative size, from the terrestrial planets to the Kuiper belt objects. Credit: CactiStaccingCrane (Wikimedia Commons, CC BY-SA 4.0); NASA/ESA/ISRO.

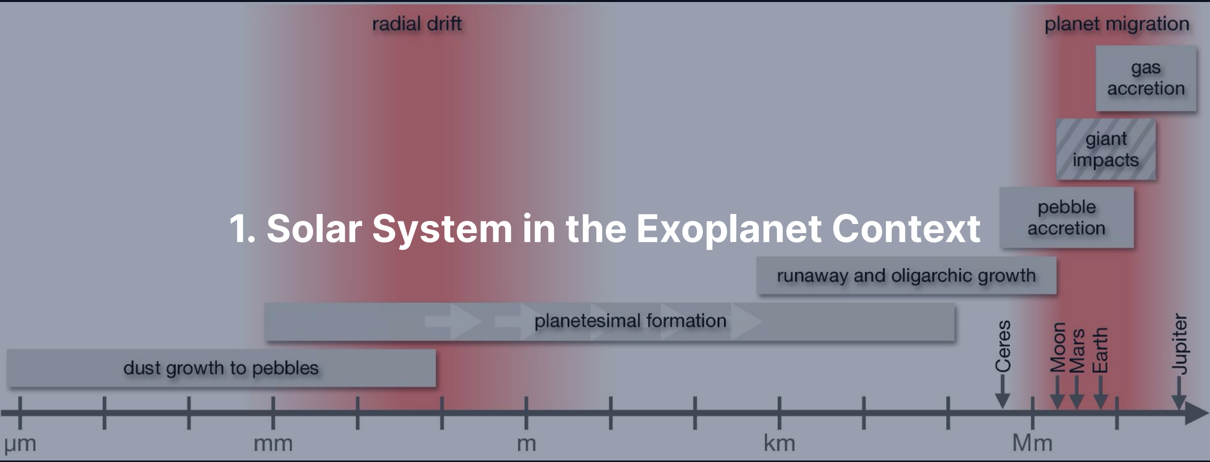
# The Thread of the Course

- L1–L8: accretion, differentiation, internal heat engines, and atmosphere-surface coupling
- L9–L12: terrestrial and giant planet evolution traced body by body
- L13: over 6000 exoplanets revealing architectural diversity

**Today:** Step back to evaluate the solar system in context and test astrobiological habitability.

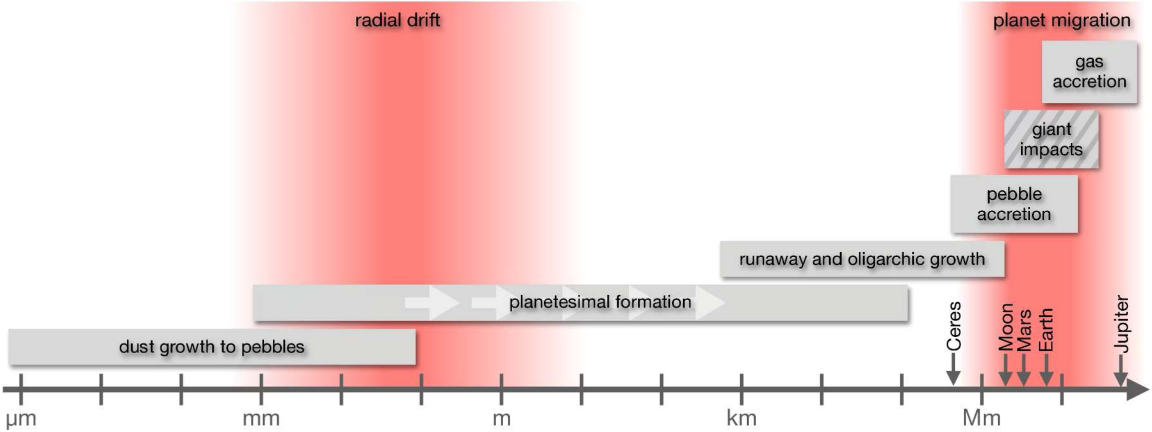



Architecture of the solar system on a logarithmic semi-major axis scale with symbol area scaled to radius; shaded spans mark the asteroid belt, the Kuiper belt and the water snow line region. Credit: Course-original figure; NASA NSSDC.



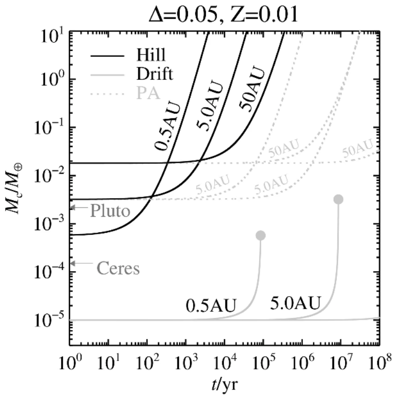

# 1. Solar System in the Exoplanet Context

Dust-to-planet growth. Drążkowska et al. (2023)



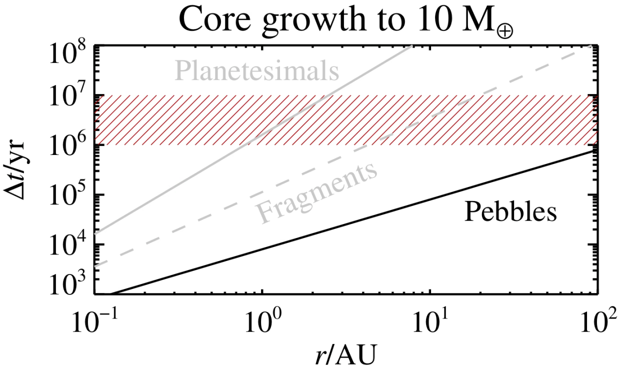

Growth processes from dust to planets. Reproduced from Drążkowska et al. (2023).

Dust grows to pebbles, planetesimals and embryos by streaming instability, pebble accretion and runaway envelope accretion; the disk lifetime of 3–5 Myr is the deadline, and ALMA disk substructure is direct observational input.



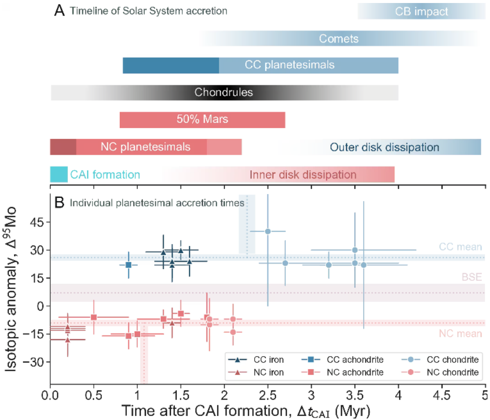

Core mass against time at 0.5, 5 and 50 AU: pebble accretion (solid) against classical planetesimal accretion (grey dotted), with Ceres and Pluto marked. Reproduced from Lambrechts & Johansen (2012).

Pebble accretion reaches 10 Earth masses well within a typical disk lifetime; planetesimal accretion at 5 AU takes longer than the disk lives.



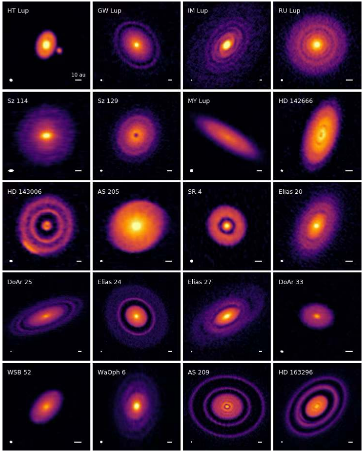

Time to grow a 10 Earth mass core against distance from the star by pebble accretion (black), planetesimal accretion (grey) and fragment accretion (dashed grey); the hatched band is the 1 to 10 Myr gas dispersal. Reproduced from Lambrechts & Johansen (2012).

A core that is to capture an envelope must reach 10 Earth masses before the gas goes: the pebble line stays below the band at every radius, the planetesimal line rises through it within a few AU.



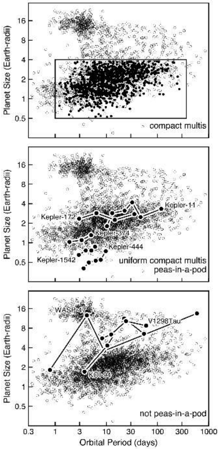

Timeline of solar system formation from isotopic dating of meteorites: the non-carbonaceous (NC, red) and carbonaceous (CC, blue) reservoirs. Reproduced from Lichtenberg et al. (2023).

The two reservoirs keep distinct isotopic signatures from CAI formation onward, so they accreted in physically separated regions of the disk for at least the first 2 to 3 Myr.



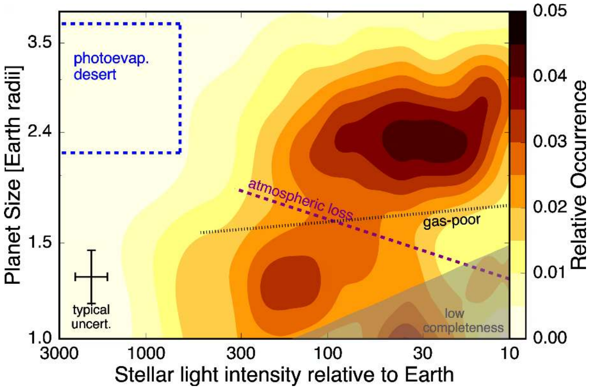

ALMA DSHARP 240 GHz continuum images of 20 nearby disks. Reproduced from Andrews et al. (2018).

Concentric rings, gaps and asymmetries are nearly ubiquitous, and most are interpreted as signatures of planet growth in progress.




Transiting census with compact multi-planet systems boxed: five uniform ones (middle), three non-uniform ones (bottom). Weiss et al. (2023, PPVII).

Many compact multi-planet systems show striking size uniformity (middle), while others do not (bottom).

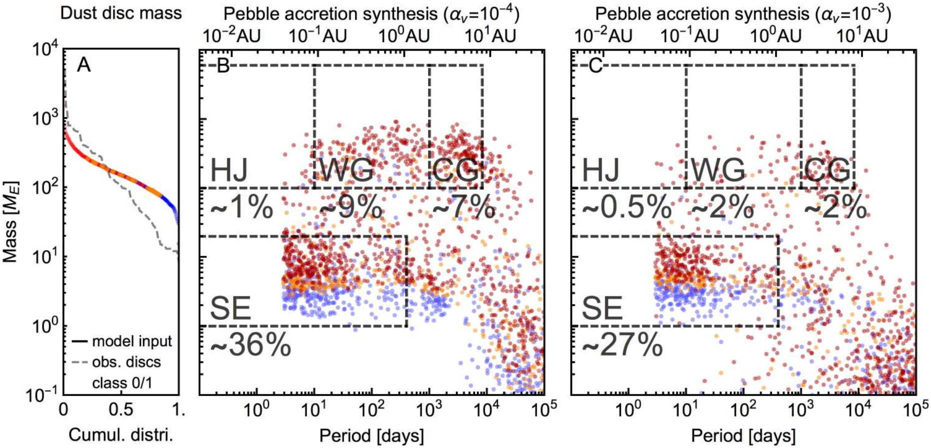



The Fulton gap: planet size against incident stellar flux for small close-in exoplanets, with the deficit near 1.8 $R_\oplus$ between super-Earths and sub-Neptunes; the dashed line marks the atmospheric-loss model, the dotted line the gas-poor model. Reproduced from Fulton et al. (2017).

Photoevaporation models reproduce both the location and the slope of the valley with irradiation.

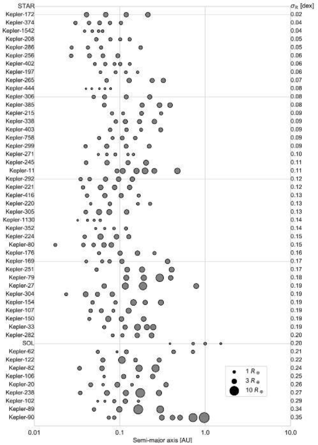
# Where the Solar System is Atypical

- No super-Earths or sub-Neptunes: the most common exoplanet size class is absent
- No close-in giant planets: hot Jupiters orbit ~ 0.5–1% of Sun-like stars, hot Neptunes are rarer still
- Outer giants on wide, circular orbits, while the inner terrestrials lack compact uniformity



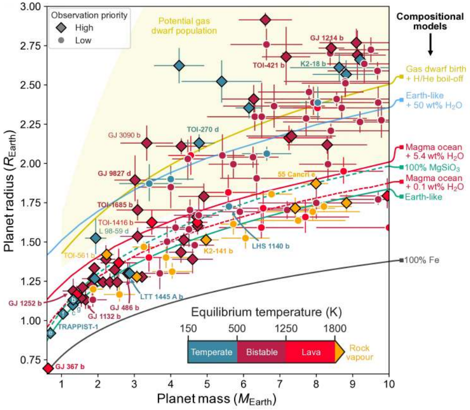

Population synthesis predictions for planet mass against orbital period under pebble accretion at low (B) and moderate (C) disk turbulence. Credit: Drążkowska et al. (2023).

At either turbulence level, pebble accretion readily forms the super-Earths and close-in giants that the solar system lacks.



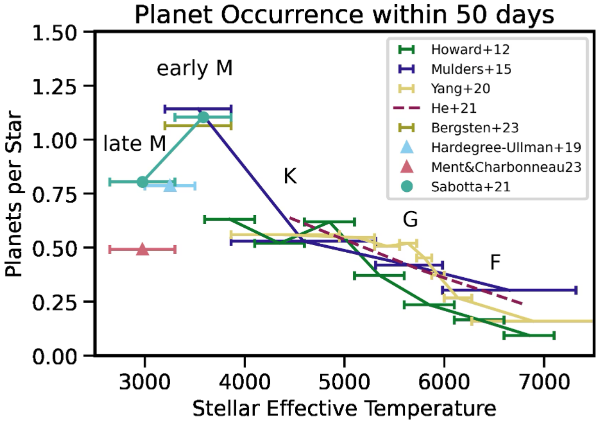

Compact multi-systems with at least four transiting planets within 1.52 AU, ranked by the uniformity of their planet sizes. Reproduced from Weiss et al. (2023).

The solar system falls in the bottom quintile of size uniformity: the inner solar system did not produce the typical compact-multi architecture.




Mass-radius diagram with composition tracks: pure Fe, Earth-like, $MgSiO_3$, water-rich, magma plus $H_2O$, and gas dwarf. Lichtenberg et al. (2025).

The close-in super-Earths sit on the Earth-like curve as stripped cores that lost their gas envelopes; larger planets carry volatile envelopes that swell their radii.



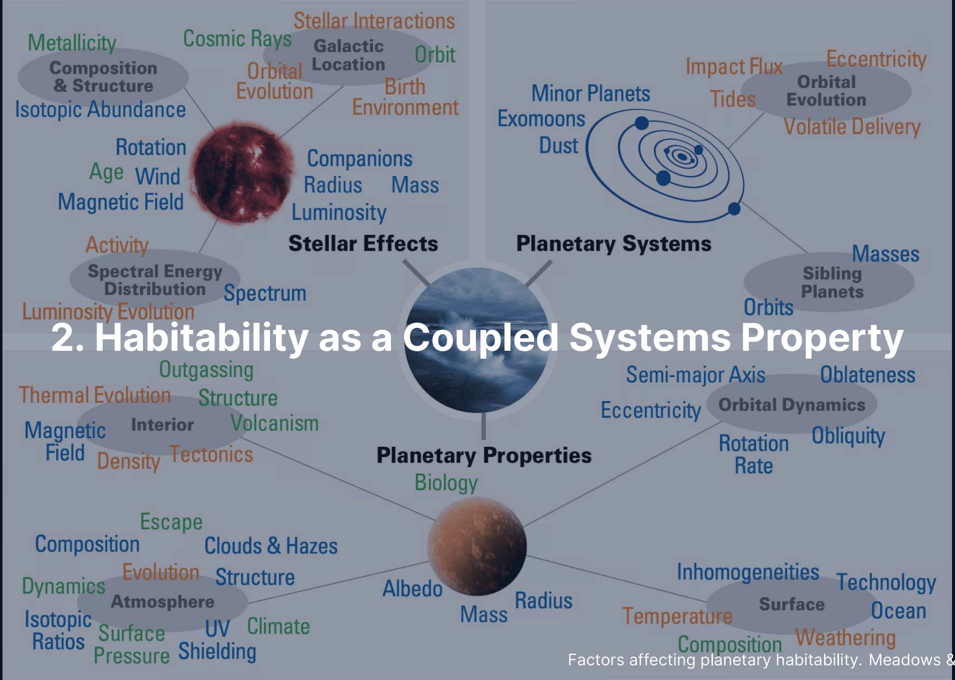

Occurrence of small planets (1–4 $R_\oplus$, $P$ < 50 d) per star against stellar effective temperature. Mulders (2024).

Occurrence is roughly twice as high around early M dwarfs as around F and G dwarfs and drops again for the late M dwarfs, where the samples are small: small planets are most common around low-mass stars.

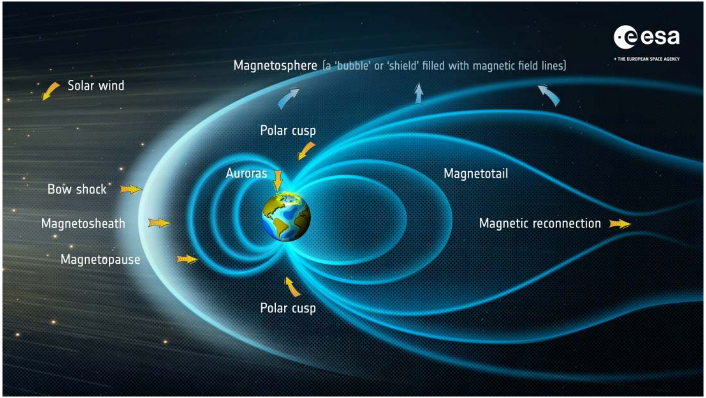
The Milky Way over the ELT site. Credit: C. Letelier/ESO.

Whether our architecture is rare stays open until PLATO (2027) and Gaia DR4 (late 2026) reach the 1 AU analogues; the 39 m ELT will then image temperate rocky planets around nearby M dwarfs directly.

# 2. Habitability as a Coupled Systems Property

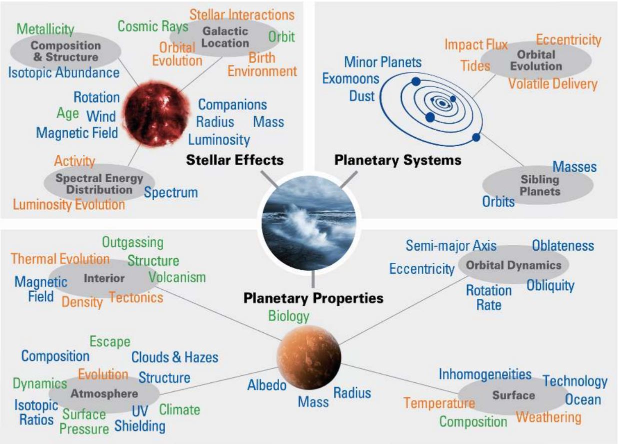



Factors affecting planetary habitability. Meadows & Barnes (2018)



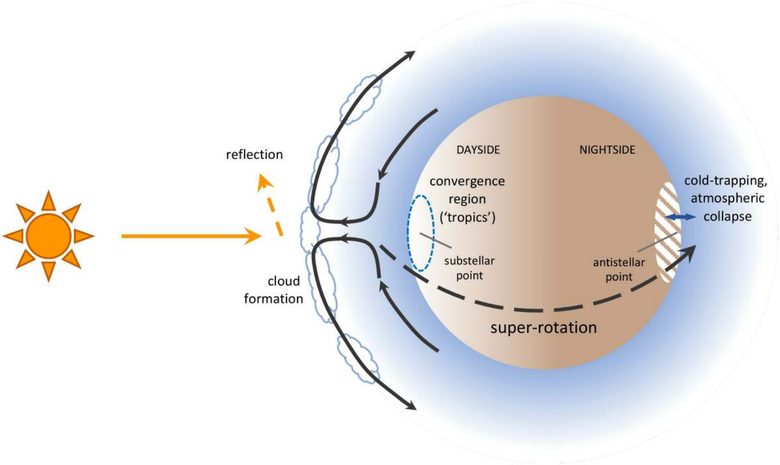

Anatomy of Earth's magnetosphere, one output of the geophysical engine. Credit: ESA, CC BY-SA 3.0 IGO.

Habitability is a coupled stack, not a checklist: a stable, benign star and orbit; a geophysical engine that drives a dynamo and recycles volatiles; an atmosphere that keeps surface water liquid; a biosphere that feeds back on weathering and redox.



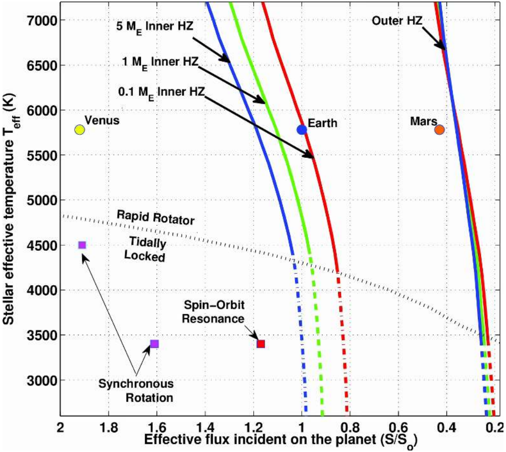

Factors affecting habitability: stellar, planetary and orbital properties jointly shape the surface environment; blue observable, green model-interpreted, orange theoretical. Meadows & Barnes (2018).

Breaking any one of these couplings can end surface water stability.



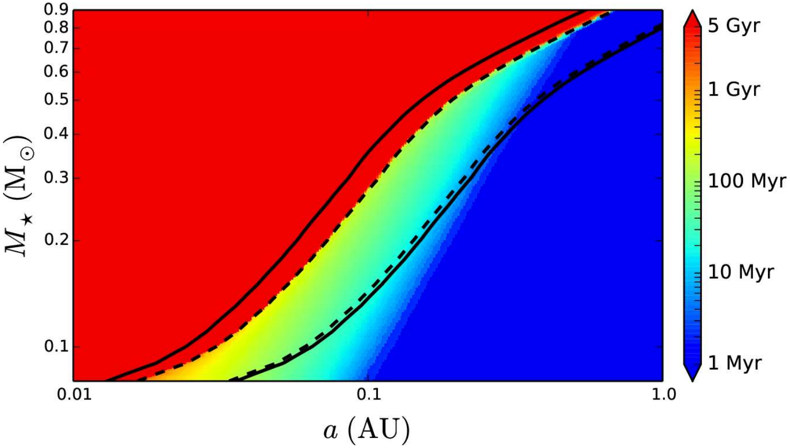

Atmospheric circulation on a tidally locked rocky planet: substellar convection and equatorial super-rotation transport heat. Wordsworth & Kreidberg (2022).

Nightside cold traps risk atmospheric collapse.




Habitable zone limits against stellar effective temperature. Kopparapu et al. (2014).

Inner edge: the runaway greenhouse limit; outer edge: the maximum $CO_2$ greenhouse; stars brighten over their main-sequence lifetime, so orbiting in the zone today does not guarantee a habitable history.




Duration of the runaway greenhouse phase on a water-bearing planet against stellar mass.
Luger & Barnes (2015).

Planets around late M dwarfs spend up to a Gyr *inside* the runaway zone during the star's bright pre-main sequence, and lose their initial water inventory.

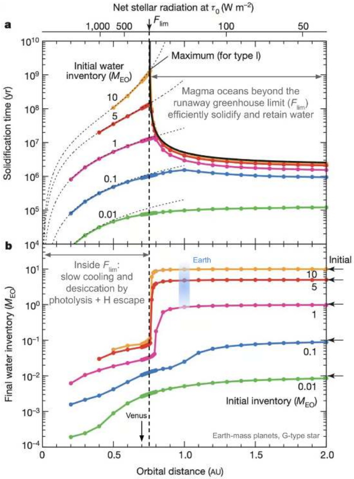
# 3. Magma Oceans, Water Delivery, Tectonic Thermostat


Magma-ocean differentiation. Lichtenberg et al. (2023)

## Why ”Delivery” Isn’t the Right Frame

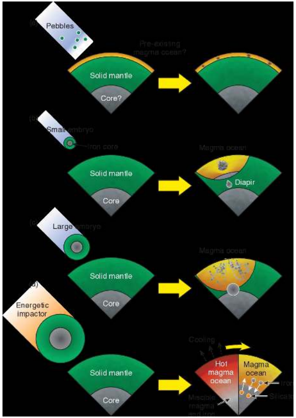
- Earth, Venus, and Mars received similar initial accreted volatile budgets
- Magma ocean partitioning sets the core, mantle, and steam atmosphere reservoirs
- Evolutionary controls: early hydrodynamic escape, mantle outgassing, and stellar brightening

Terrestrial volatile inventories diverge because differentiation and evolution matter far more than initial delivery.



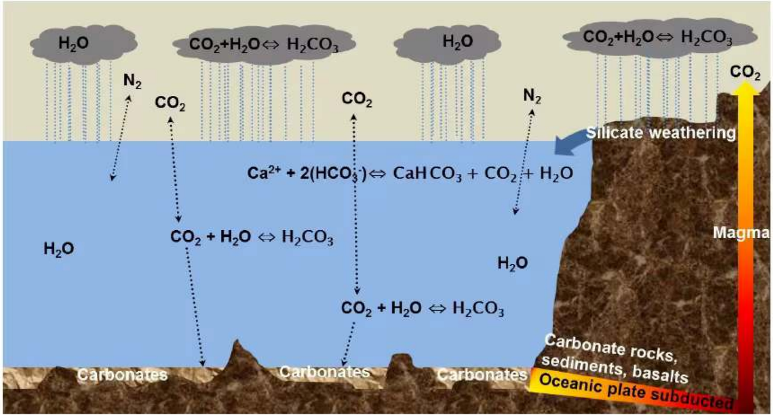

Two types of terrestrial planet from magma ocean evolution. Credit: Hamano et al. (2013).

Inside the critical flux, Type II planets lose their water to hydrodynamic escape before solidifying; Type I planets further out solidify quickly and retain oceans.



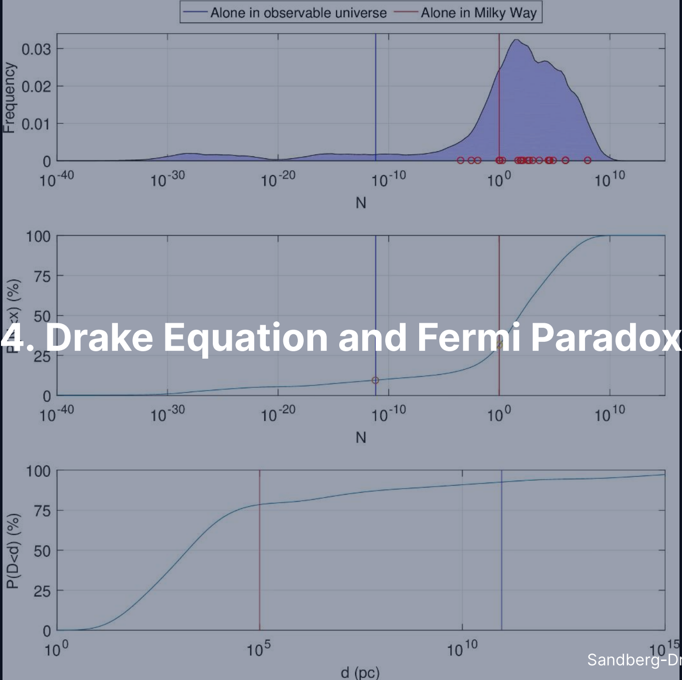

Magma ocean differentiation into a metal core, a silicate mantle and a steam atmosphere. Lichtenberg et al. (2023).

Volatiles partition between liquid silicate and the steam atmosphere; the sequestered fraction depends on mantle redox, solidification time and depth, so identical initial budgets yield radically different atmospheric masses.




The carbonate-silicate cycle. Credit: Lammer et al. (2018).

Volcanic $CO_2$ outgassing is balanced by temperature-dependent silicate weathering and subduction, a negative feedback that needs surface liquid water, active volcanism and tectonic recycling; its failure produces the Venus-like 92 bar.




# 4. Drake Equation and Fermi Paradox

Sandberg-Drexler-Ord Drake posterior (2018)

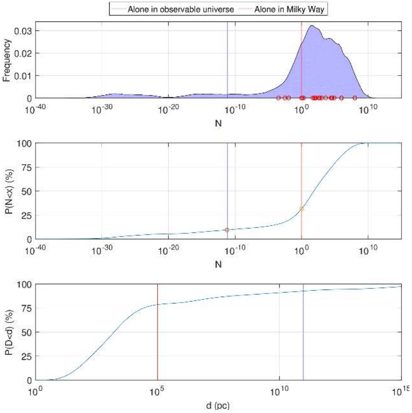
# The Drake Equation: A Framing Tool

$$N = R_\star \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

$R_\star$ star-formation rate, $f_p$ fraction with planets, $n_e$ habitable planets per system, $f_l$ life, $f_i$ intelligence, $f_c$ communication, $L$ lifetime of a signalling civilisation.

- Astrophysical factors are now measured: $R_\star$, $f_p \sim 1$, $n_e \approx 0.1$–$0.6$ (Bryson+ 2021)
- Biological and social terms ($f_l, f_i, f_c, L$) span 4–10 orders of magnitude
- Factors co-evolve rather than acting as independent pipeline stages

The Drake equation is a tool for framing ignorance, not a numerical estimator.




Monte Carlo posterior for the number $N$ of communicating civilisations, sampled over the full published parameter ranges. Sandberg, Drexler & Ord (2018).

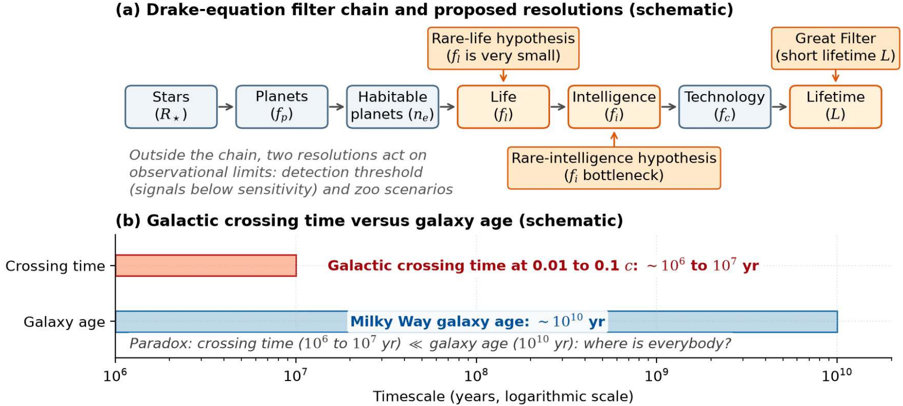
The distribution is broad and heavy-tailed, with about a third of the probability below $N = 1$ in the Milky Way: the apparent cosmic silence is unsurprising once the priors are honest.

# Why So Broad? Log-Sum of Broad Priors

Taking the logarithm turns the Drake product into a sum of broad priors:

$$\log_{10} N = \log_{10}(R_\star f_p n_e) + \log_{10} f_l + \log_{10} f_i + \log_{10} f_c + \log_{10} L$$

- Unconstrained factors have log-uniform priors spanning many decades
- The log-sum of broad, skewed priors spreads $\log N$ over tens of decades; the long low tail of $f_l$ puts a third of the mass below $N = 1$
- The posterior spans ~ 30 decades, placing high probability on $N < 1$




The Fermi paradox as a chain of filters, with a galactic crossing of $10^6$ to $10^7$ yr against a $10^{10}$ yr galaxy age. Course figure.

The silence is an intuition check on subjective priors, not physical evidence: rare abiogenesis or intelligence, a Great Filter that keeps $L$ short, or transmissions below current detection limits.

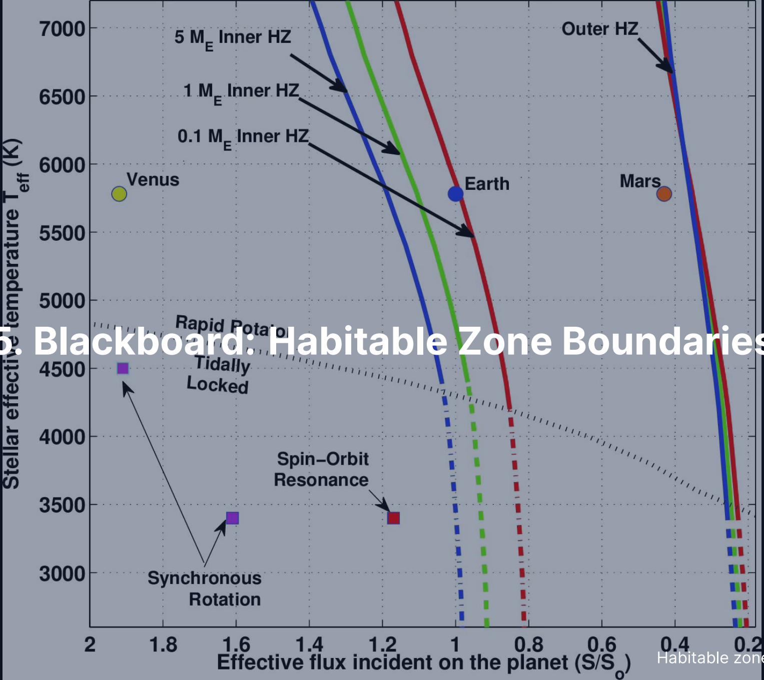
# Break

End of Part 1: Solar System Context and Habitability

Part 2: Habitable Zone Derivation, Astrobiology, Life Detection

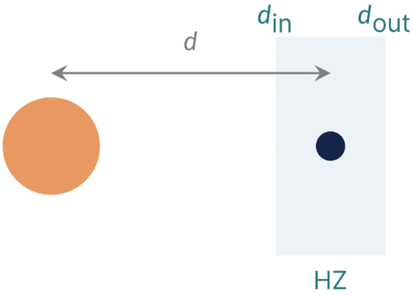
Earthrise, Apollo 8 (1968). Credit: NASA

# 5. Blackboard: Habitable Zone Boundaries




Habitable zone vs $T_{\rm eff}$. Kopparapu (2014)

# Goal and strategy




**Goal:** locate the inner and outer edges of the habitable zone from one energy balance.

- Inverse square in, Stefan-Boltzmann out: $T_{\mathrm{eq}}(d)$
- Express the boundaries as an effective flux $S_{\mathrm{eff}}$
- Insert the runaway limit (Lecture 9) and the maximum $CO_2$ greenhouse

Star $L_\star$, planet at $d$ with Bond albedo $A_B$: the zone is the strip between $d_{\mathrm{in}}$ and $d_{\mathrm{out}}$.

# Setup: Stellar Flux and Equilibrium Temperature

Star with bolometric luminosity $L_\star$, planet at distance $d$:

$$F_\star(d) = \frac{L_\star}{4\pi d^2}$$

Bond albedo $A_B$ (reflected fraction), rapidly rotating sphere, equilibrium:

$$T_{\mathrm{eq}}(d) = \left[\frac{L_\star(1 - A_B)}{16\pi\sigma\epsilon\, d^2}\right]^{1/4}$$

For Earth: $A_B = 0.3$, $\epsilon = 1$, $L_\star = L_\odot$, $d = 1$ AU $\rightarrow T_{\mathrm{eq}} \approx 255$ K; the surface is 288 K from ~ 33 K greenhouse warming. *To the board.*

# Result: Habitable Zone Boundaries

Effective flux $S_{\mathrm{eff}} = F_\star(d)/S_\oplus$ and the boundary distances:

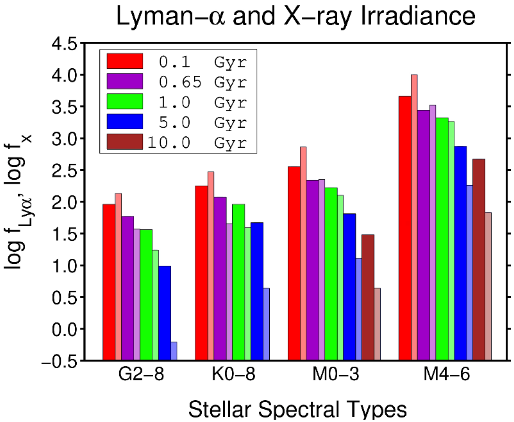
$$d = 1\ \mathrm{AU}\sqrt{\frac{L_\star/L_\odot}{S_{\mathrm{eff}}}}$$

- Inner edge: runaway greenhouse (Simpson-Nakajima, Lecture 9), $S_{\mathrm{in}} \approx 1.06$, $\approx 0.97$ AU for the Sun
- Outer edge: maximum $\mathrm{CO_2}$ greenhouse, $S_{\mathrm{out}} \approx 0.35$, ~ 1.69 AU for the Sun
- Main-sequence lifetime $t_{\mathrm{MS}} \approx 10\ \mathrm{Gyr}\,(M_\star/M_\odot)^{-2.5}$ sets how long the zone lasts

# HZ Across Stellar Types

Habitable zone distance scales with stellar luminosity: $d \propto \sqrt{L_\star}$.

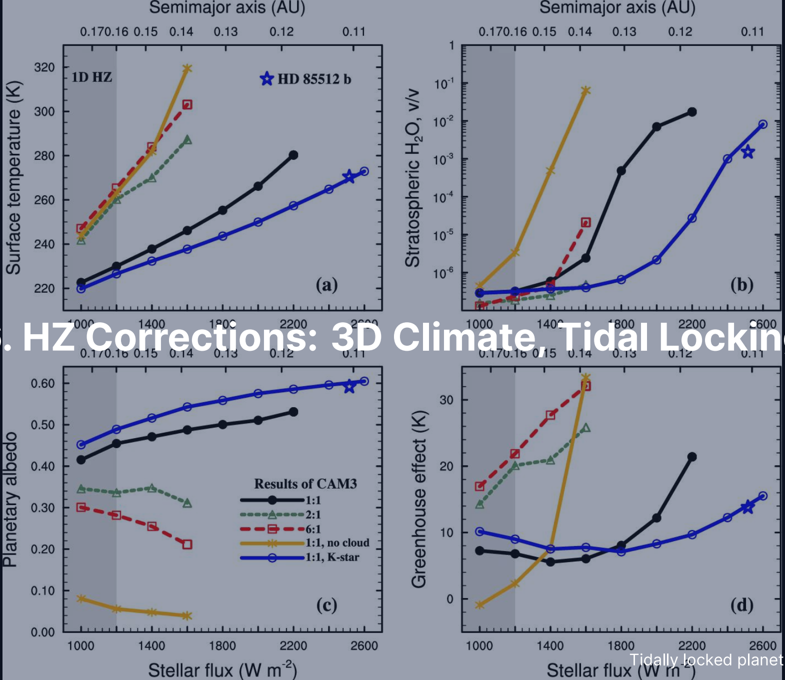
- **G dwarf** ($L_\odot$): $d$ = 0.97–1.69 AU; main-sequence lifetime ~ 10 Gyr
- **K dwarf** ($0.3\,L_\odot$): $d$ = 0.53–0.93 AU; lifetime 24 Gyr (orange-dwarf sweet spot)
- **M dwarf** ($5 \times 10^{-4}\,L_\odot$): $d$ = 0.022–0.038 AU; lifetime $10^{12}$ yr, but tidally locked with a long pre-MS runaway



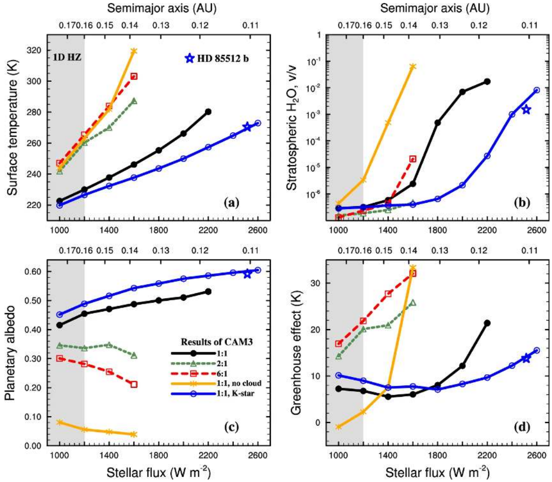

Lyman-$\alpha$ and X-ray fluxes at the Earth-equivalent distance against stellar type and age. Cuntz & Guinan (2016).

M dwarfs deliver over 100 to 500 times the G-dwarf flux; X-rays drop 500-fold over 5 Gyr but Lyman-$\alpha$ only 5 to 10-fold, so early K dwarfs offer long lifetimes without the M-dwarf XUV extremes.

# 6. HZ Corrections: 3D Climate, Tidal Locking

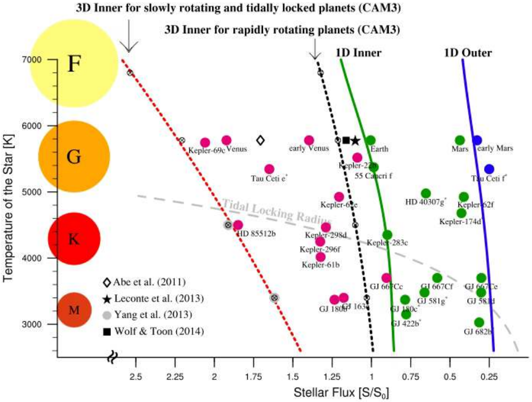



Tidally locked planet clouds. Yang et al. (2013)




3D global climate model of a tidally locked planet with clouds. Yang et al. (2013).

Persistent substellar convection lifts thick water clouds, the planetary albedo rises to ~ 0.6, and the inner edge tolerates ~ 2× the flux that 1D models predict.




Habitable-zone limits for M dwarfs with stellar flux increasing to the left: 1D Kopparapu boundaries (solid), the 3D inner edge (red dotted) and the tidal-locking radius (grey dashed). Shields et al. (2016).

The 3D inner edge tolerates 1.7–2.2× the 1D flux, and the tidal-locking radius encloses almost all M-dwarf habitable-zone planets.

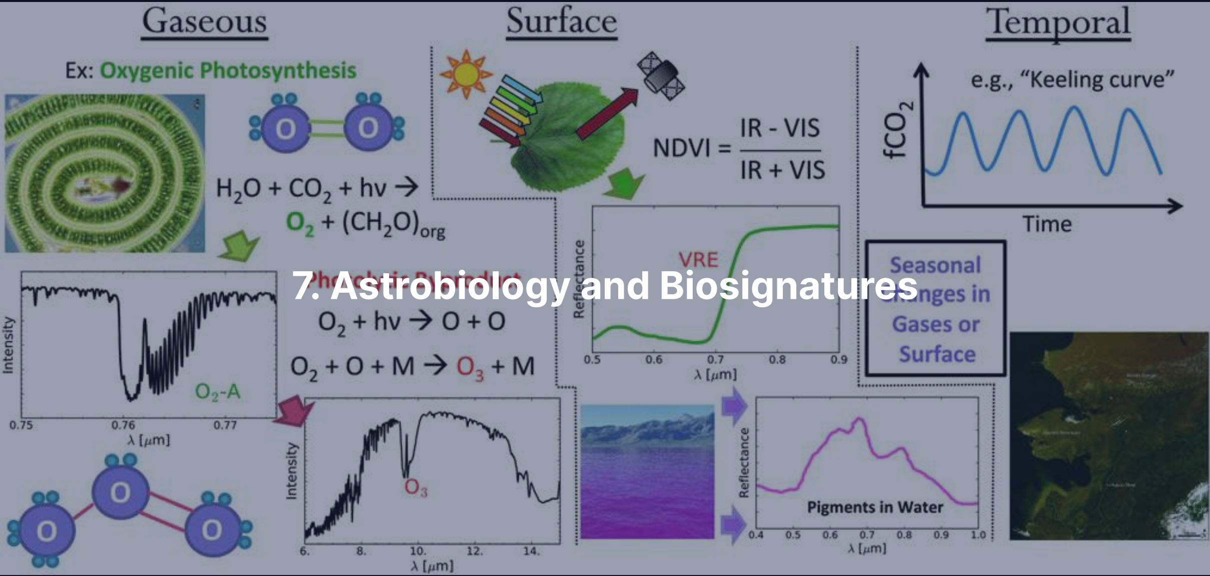
# Tidal-Locking Timescale

Tidal despinning scales steeply as $a^6$, so close-in habitable planets around low-mass stars synchronise quickly:

$$\tau_{\text{lock}} \sim \frac{\omega_0\, a^6\, I_p\, Q_p}{3\, G\, M_\star^2\, k_{2,p}\, R_p^5}$$

- Earth at 1 AU has $\tau_{\text{lock}} \sim 10^{12}$ yr, far exceeding the age of the universe
- TRAPPIST-1 b ($a \approx 0.011$ AU) synchronises in $\lesssim 10^7$ yr
- With a system age of ~ 8 Gyr, all known M-dwarf HZ worlds are locked

# 7. Astrobiology and Biosignatures

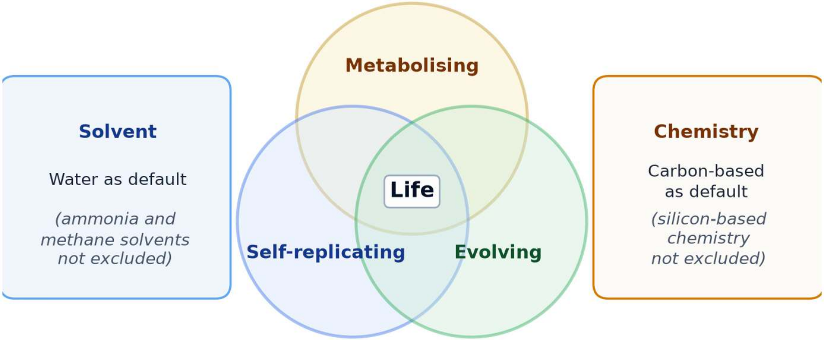



Biosignature classes. Schwieterman et al. (2018)

# Origin-of-Life Hypotheses

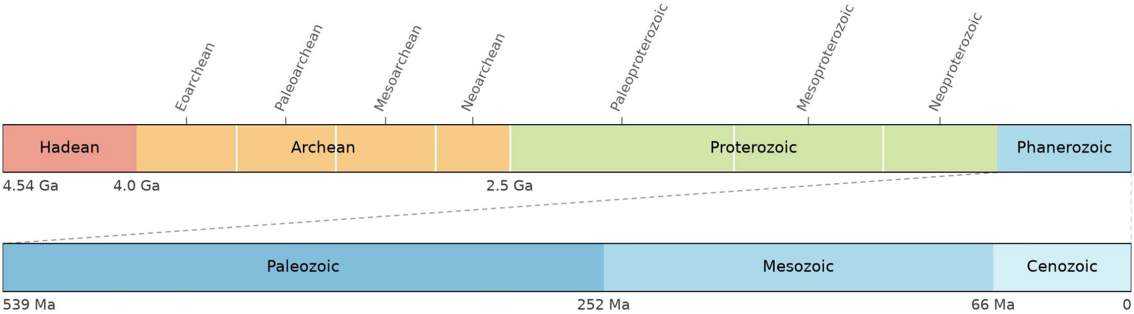
- **RNA world**: catalytic ribozymes carry the genetic code before proteins
- **Alkaline hydrothermal vents**: geochemical proton gradients drive early metabolism
- **Warm surface ponds**, where wet-dry cycles polymerise organic precursors, and **panspermia**, the transfer of viable organisms between bodies

## Working definition of life (schematic)

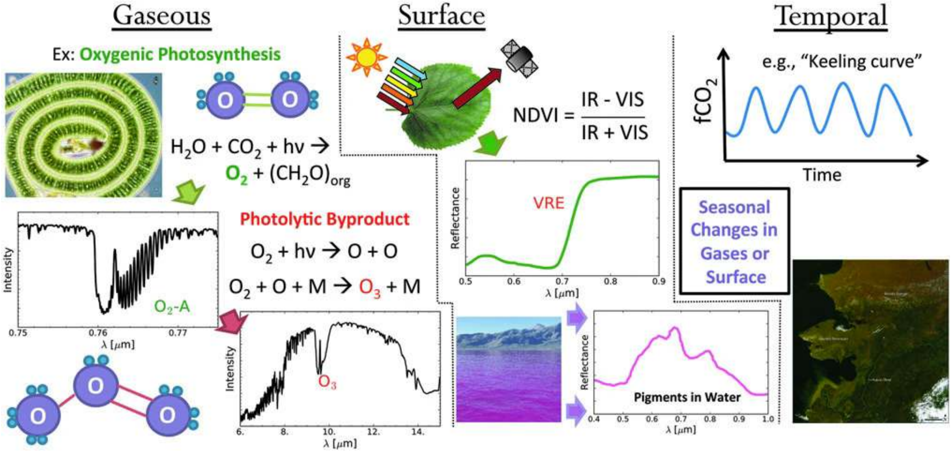



Working definition of life, with water and carbon as defaults and other solvents and chemistries not excluded. Course figure.

**Life (Lederberg 1965): a self-replicating, metabolising, evolving chemical system.**

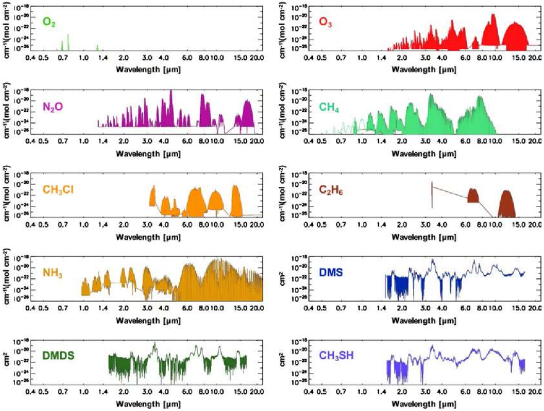



Geologic eons and eras of Earth across 4.54 Gyr: the Precambrian occupies almost 90% of the timeline, with life emerging early in the Archean. Credit: Course-original figure; Gradstein et al. (2020).




Three classes of biosignature. Schwieterman et al. (2018).

**Gaseous** ($O_2$, $O_3$, $CH_4$, $N_2O$, organosulfur species), **surface** (the vegetation red edge near 700 nm) and **temporal** (seasonal cycles such as Earth's $CO_2$ Keeling oscillation): a credible detection needs several classes together.



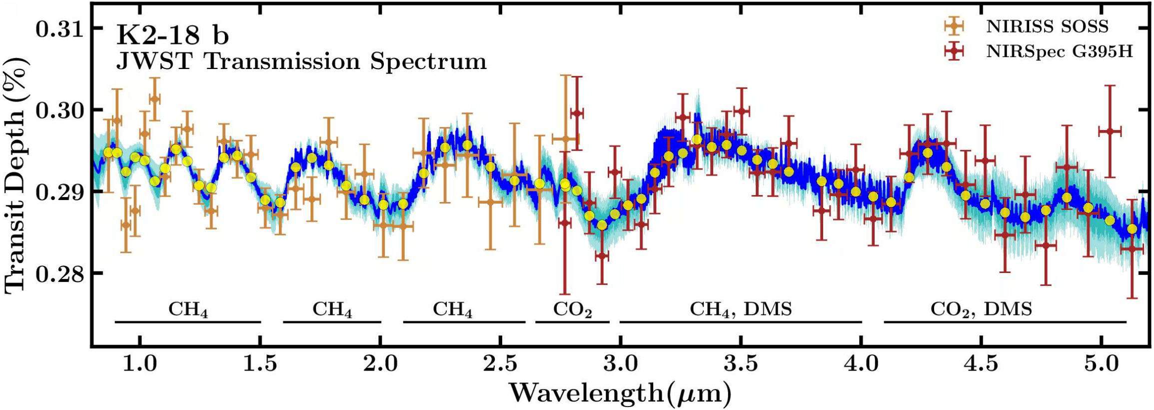

Absorption bands from 0.4 to 20 $\mu$m of ten candidate biosignature gases. Schwieterman et al. (2018).

$O_3$ at 9.6 $\mu$m, $CH_4$ at 3.3 and 7.7 $\mu$m, $N_2O$ and the DMS and DMDS window at 6–15 $\mu$m set the needs of the mid-IR interferometer LIFE; $O_2$ at 0.76 $\mu$m and the visible $O_3$ bands set those of the coronagraph HWO.

# Disequilibrium is the Diagnostic

- Individual gases in isolation almost always possess plausible abiotic origins
- Earth pair: $O_2$ (21%) and $CH_4$ (1.8 ppm) react on decade timescales
- Simultaneous persistence requires massive, continuous biological resupply fluxes

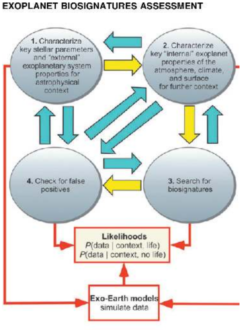
Chemically incompatible gases coexisting in an atmosphere are the most convincing remote signature of active biology.

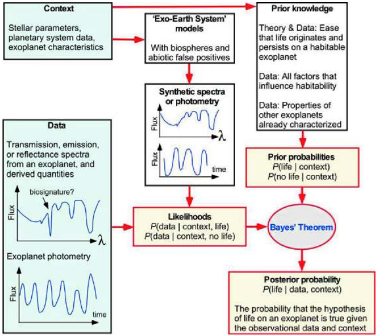



JWST transmission spectrum of K2-18 b combining NIRISS and NIRSpec observations. Credit: Madhusudhan et al. (2023).

Absorption bands reveal atmospheric $CH_4$ and $CO_2$ alongside a tentative DMS feature near 3.4 $\mu$m.

# Catling 2018: A Bayesian Framework




Detection is a Bayesian chain of inference over stellar context, planetary properties and abiotic mechanisms, not a single yes/no spectral peak.

Source: Catling et al. (2018)

# 8. False Positives and Solar System Targets

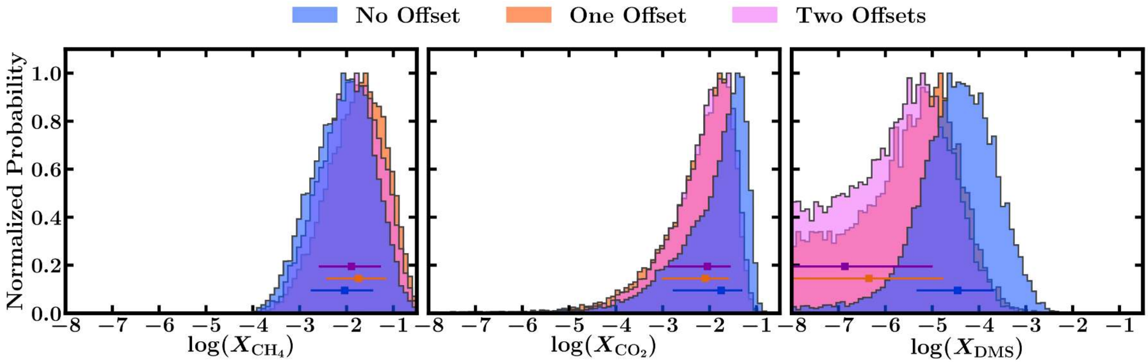


Venus phosphine controversy. Lincowski et al. (2021)

# False Positives: Every Gas Has an Abiotic Pathway

- Abiotic $O_2$: $H_2O$ photolysis followed by hydrogen escape, or $CO_2$ photolysis
- Abiotic $CH_4$: hydrothermal serpentinisation or volcanic outgassing at low oxygen fugacity
- Abiotic $N_2O$ from lightning and photochemistry; DMS has no significant abiotic Earth source, but one could exist elsewhere

Verification is an inverse problem: confirmation requires ruling out every plausible geochemical and photochemical false positive.



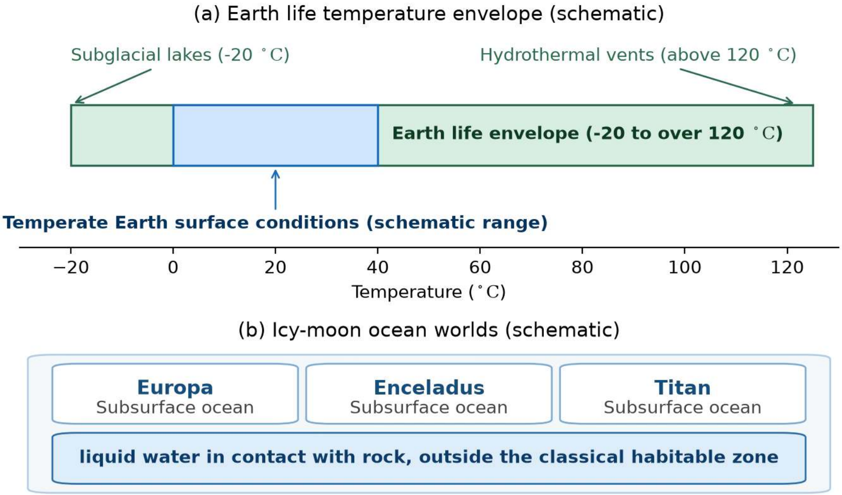

Retrieved mixing ratio posteriors for $CH_4$, $CO_2$ and DMS in K2-18 b under zero, one and two floating instrument offsets. Credit: Madhusudhan et al. (2023).

$CH_4$ and $CO_2$ hold under every offset choice, while the DMS significance collapses from 2.4$\sigma$ to below 1$\sigma$.

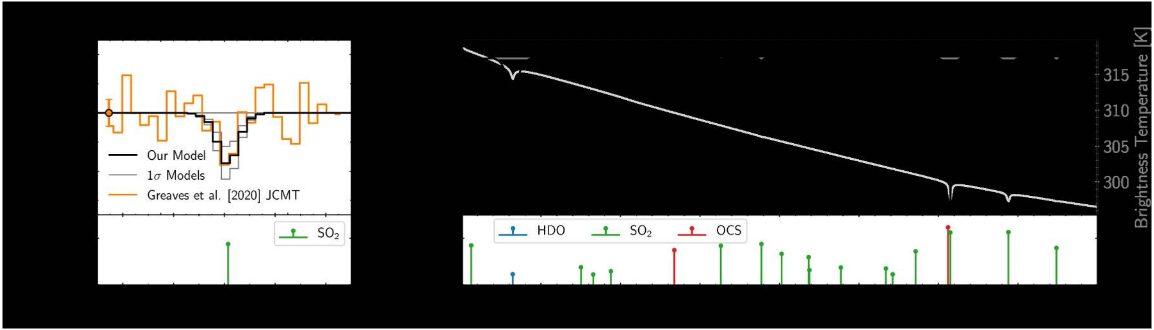
# Solar System Targets for Life Detection

- Mars: ancient habitable lacustrine environments; the Mars Sample Return path
- Europa and Enceladus: confirmed subsurface oceans; Enceladus's plume carries $\mathrm{H}_2$, organics and phosphates from hydrothermal activity, hypothesised for Europa
- Titan and Venus: complex hydrocarbon hydrology and contested cloud chemistry

Five prime environments with liquid water reservoirs or rich organic chemistry.



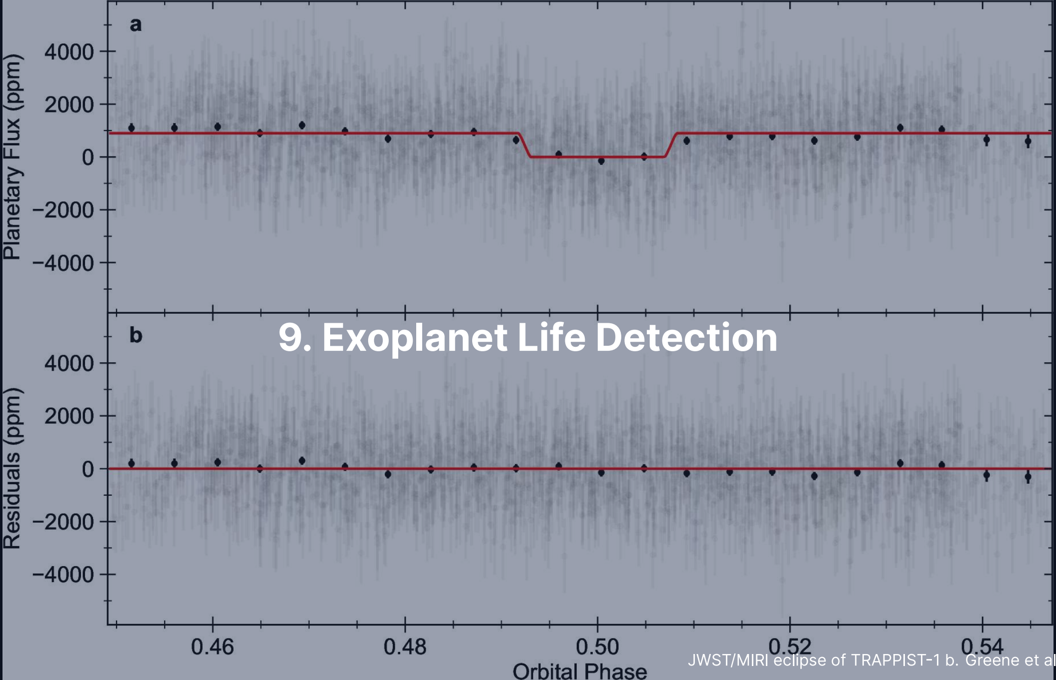

Extremophiles widen the definition of habitable: Earth life spans −20 to over 120°C, and the icy moons hold liquid water beneath ice outside the classical habitable zone, in contact with rock at Europa and Enceladus. Course figure.



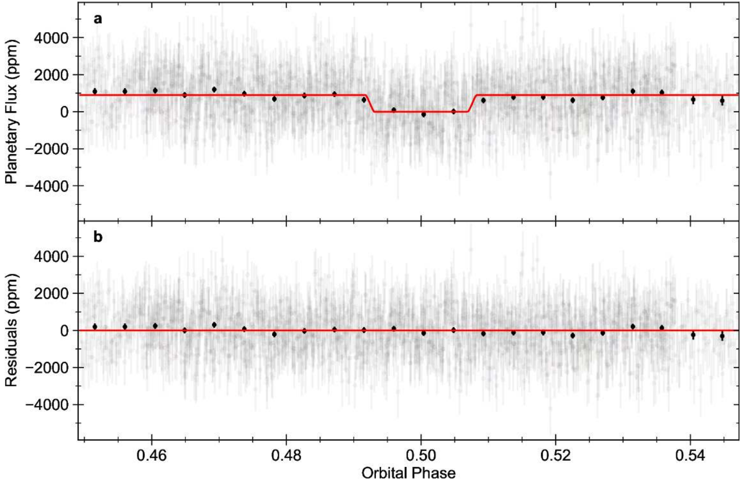

The Venus phosphine controversy: the claimed $PH_3$ line against the mesospheric $SO_2$ fit. Lincowski et al. (2021).

Greaves et al. (2021) claimed $PH_3$ in the Venus clouds, but the line is fit by mesospheric $SO_2$. DAVINCI tests Venus in situ, EnVision and VERITAS from orbit.

# 9. Exoplanet Life Detection

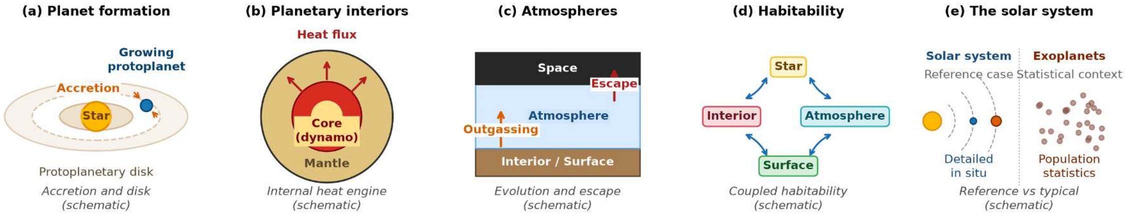



JWST/MIRI eclipse of TRAPPIST-1 b. Greene et al. (2023)




JWST MIRI 15 $\mu$m secondary eclipse of TRAPPIST-1 b. Credit: Greene et al. (2023).

The dayside brightness temperature of 503 K matches the 508 K bare-rock prediction and rules out a thick atmosphere.




The five biggest lessons: formation, interiors as heat engines, dynamic atmospheres, habitability as a coupled property, the solar system as reference case. Course figure.

| | Open question | What is at stake, and what will decide it |
|---|---|---|
| 1 | Is the origin of life a rare accident or a generic chemical inevitability? | The largest uncertainty in quantitative habitability estimates |
| 2 | What carves the radius valley between super-Earths and sub-Neptunes? | Photoevaporation, core-powered mass loss, or distinct formation channels |
| 3 | When did Jupiter form? | Architect of the inner solar system: the NC-CC dichotomy and Earth-like system rarity |
| 4 | Was Mars ever inhabited? | Mars Sample Return, the most direct test; it recalibrates the prior on $f_l$ |
| 5 | Is the solar system rare or typical? | PLATO, Gaia DR4/DR5 and long-baseline RV give a probabilistic answer |

The five biggest open questions, each with what is at stake or the test that will decide it. Course figure.

# Summary

1. Planet formation and differentiation are universal physical processes; interiors are heat engines that drive tectonics, dynamos and outgassing over Gyr
2. Atmospheres evolve through escape, outgassing and photochemistry, and habitability is a coupled trajectory, not a static orbital condition
3. The solar system is one detailed reference in a diverse exoplanet population

**Next:** The final exam.

# Thank You

Planetary Systems is complete.

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`